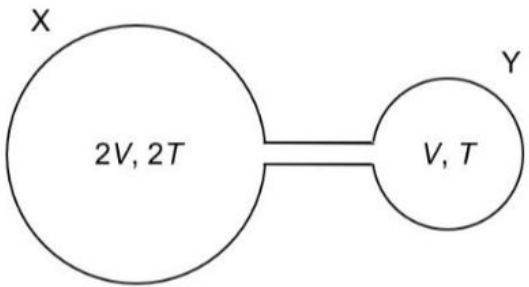
$\frac{2}{2}$ $\frac{2}{2}$ $2T$ $c_{rms,\ new}$

15 An ideal gas is contained in two spherical containers X and Y of volume $2V$ and V respectively, connected by a hollow tube of negligible volume. The containers X and Y are maintained at temperatures $2T$ and T respectively. The setup is shown in the figure below.



Determine the ratio

$$\frac{\text{number of moles of gas in container X}}{\text{number of moles of gas in container Y}}.$$

A 0.25 **B** 1.00 **C** 2.00 **D** 4.00

Ans: A

Since the containers are connected, the gas reaches the same pressure P throughout the system at equilibrium.

Using the ideal gas law: $pV = nRT$, so: $n = \frac{pV}{RT}$

Moles in Container X: $n_X = \frac{p\times 2V}{R\times 2T} = \frac{2pV}{2RT} = \frac{pV}{RT}$; Moles in Container Y: $n_Y = \frac{p\times V}{R\times T} = \frac{pV}{RT}$

Ratio: $\frac{n_X}{n_Y} = \frac{\frac{pV}{RT}}{\frac{pV}{RT}} = 1.00$